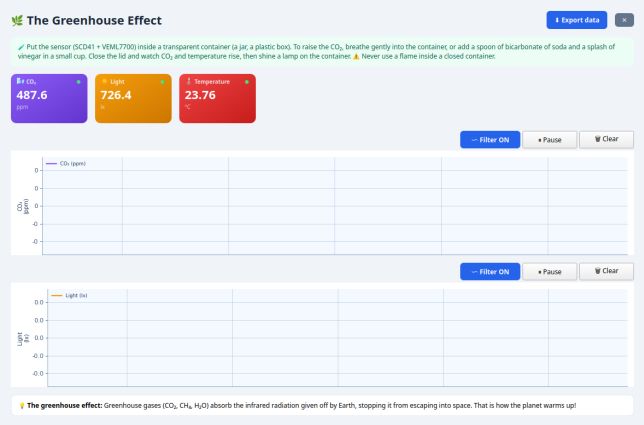
The Greenhouse Effect

A jar that warms up — and the honest question: what actually warmed it?

Level: Explorer and Scientist — ages 11–19 (Romanian school years V–XII) • **Time in class:** 20 min demonstration, or two 50-minute lessons, investigation with a control • **Sensors:** SCD41 (carbon dioxide) and the light sensor (illuminance)



The activity screen. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Explorer and Scientist — ages 11–19 (Romanian school years V–XII). As a **demonstration** it works from year V: a jar is seen to warm up. As an **honest investigation** — one that asks *what warmed it* — it needs the idea of a control and a controlled variable, so from year VII upwards. The most direct entries are in **year V** (geography: the atmosphere), **year VIII** (physics: heat transfer) and **years IX–XII** (physical geography: the radiation balance).

Assumed knowledge

The three modes of heat transfer — conduction, convection, radiation — and the difference between them. The idea of a *control*. For the quantitative version: the difference of two temperatures and a mean.

Time in class

Short: 20 min (demonstration with one jar). Full: two 50-minute lessons (1 h 40 min), **with two jars**, one of them a control — the only version from which a conclusion can be drawn.

Teacher preparation

25 min — prepare **two** identical containers and **two** identical light sources, or one lamp placed exactly midway; decide how you will raise the CO_2; and **do a ten-minute trial run with both jars empty**, to see how much they differ from each other with nothing changed.

Sensor settling time

Ten minutes after closing the containers, before you start recording. Temperature needs longer than CO_2, and the differences pursued here are small: a reading taken after two minutes means nothing.

Sensors and boxes SCD41 for CO_2 and the light sensor for illuminance; the model in the box is the VEML7700. The app also shows temperature, from the BME280, with a chart. The box runs the one firmware, `serial_wifi`, which reads the SCD41 over USB and WiFi alike; on a box still carrying the old USB-only firmware the app says to flash the WiFi one (see `firmware/FLASHING_GUIDE.pdf`). **Two identical boxes** are all but compulsory: with one there is no control, and without a control the activity cannot answer the question it asks.

Three routes

Short 20 min: demonstration — one closed container, the lamp on, the three cards watched together, and the discussion of *what* caused the warming.

Full Two 50-minute lessons: lesson 1 — the scatter of the two empty jars, and the jar with a lid against the one without; lesson 2 — two identical containers, one with raised CO_2 and one control, an identical lamp on each, ten minutes of settling and twenty-five minutes of measurement, with the temperature difference tracked between them.

Project Extension: a third container, *without a lid*, alongside the other two in lesson 2 — it shows, in the same run, how much of the warming is simply convection being stopped.

Read this before planning the lesson — this experiment does not demonstrate what it appe

The temptation is to put the box in a jar, raise the CO_2 and watch both the CO_2 and the temperature *rise*. Both really do rise. **But the temperature rise does not demonstrate the greenhouse effect**, and saying that it does is the commonest error in teaching this subject. That is why the app's panel proposes the rig with **two identical jars, one a control**, and says itself that at jar scale the CO_2 effect is usually smaller than the scatter of the set-up.

A closed jar under a lamp warms up **primarily because it has a lid**: the lid stops convection, that is, stops the warm air leaving. That is exactly how a garden greenhouse works — by stopping convection — and **exactly how the atmospheric greenhouse effect does *not* work**, being a radiative phenomenon. The name is a historical accident and it misleads.

What can be done honestly: **two identical jars**, with the same lamp, one with raised CO_2 and one control. Then the difference between them — if it exceeds the scatter of the two empty jars, measured beforehand — is the only thing attributable to the gas. And if it does not exceed it, the correct answer is *we cannot detect it*, and that is a good answer.

Section 5 sets out what this activity really does prove, and it is something better than the screen promises.

1. What the app shows and what it does not

It is an **open instrument**, and a deliberately simple one: no steps, no green button, no Δ and no comparison between two boxes. It shows three quantities and their charts, and everything else — including the question they answer — you bring yourself.

- **Three cards**: **CO_2** in `ppm`, **Temperature** in °C and **Illuminance** in `lx`.
- **Three charts over time**, one below the other: **CO_2**, **Temperature in the jar** — the quantity the activity is about — and **Illuminance**. They show **the last 30 minutes**, with the time axis in minutes and seconds.
- **An instruction panel** on the right: **How to set it up** (two identical jars with lids, under the same lamp, one box in each; CO_2 raised in one jar only, the same gesture without the gas in the control; only the box in the jar, the laptop and power bank outside; the confounds to

avoid) and a short **The greenhouse effect:** paragraph. The panel warns, correctly, never to use a flame in a closed jar.

- At the top, beside the title, **Change box:** and the **Export data** button.

Technical note — what is on screen, what is missing, and how to compensate

The sensors. SCD41 for CO_2 (the `CO2` column), the light sensor for illuminance (the `LUX` column) and BME280 for temperature (the `T` column).
How it starts. From the **Activities** page, from the **The Greenhouse Effect** card, with the **Start** button. A box is chosen when it opens, so **for two jars you open the app twice**, one window per box. Name the boxes in **Settings** (for example "Control" and "CO2"): the name appears in the window and in the exported files, and each box keeps its own colour on the charts.
Temperature has its own chart — and it is the quantity that matters. The charts keep the last 30 minutes, so a whole measuring run fits on screen. Even so, **record the values on paper** — on the sheet, every ten minutes, or every minute on a separate page for the quantitative version: comparing two windows is safer on paper than by eye, and the exported file keeps the whole session.
The electronics stay outside. Only the box goes into the jar, its cable passed under the lid; the laptop, the power bank and anything else that warms up stay outside. The box itself warms a little (the board inside draws current), so **each jar must hold a box of the same kind, powered the same way** — otherwise you have added a heat source to one jar only.
There is no Δ and no comparison between boxes. Unlike *Air Quality* or *Urban Heat Islands*, this app compares nothing: each window shows one box. **The difference between the CO_2 jar and the control is computed by you**, from the two windows or from the two exported files.
Exporting. Export data, top right, for the box in the current window; by default an `.xlsx` file. With two boxes, export from each window separately and **give the files names that say which jar was which** — for example `control` and `co2` appended to the suggested name. Without that, in a week's time nobody will know which is which.
How long you wait. Ten minutes after closing, with the lamp off, before switching the lamp on and recording the first reading — as the app's panel also asks. The differences pursued here are small, and temperature settles far more slowly than CO_2. The CO_2 sensor gives a new value about every five seconds.

2. Where this fits in the curriculum

- **Geography** — in **year V**, the Earth's atmosphere and its role; in **years IX–XII**, the Earth's radiation balance and climate. The activity offers something rare here: the chance to distinguish a *mechanism* from an *analogy*.
- **Physics** — in **year VIII**, heat transfer by conduction, convection and radiation. **This is the most valuable entry**, because the activity's central question is precisely "*which* of the three warmed the jar?" At upper secondary, the absorption of radiation by molecules, in the atomic and molecular physics chapters (check the year in the syllabus in force).
- **Chemistry** — in **years IX–X**, the reaction between bicarbonate and vinegar, if you choose the chemical route to raising the CO_2.
- **Mathematics** — the difference of two temperature series and the idea that a difference counts only relative to the scatter of the measurement — the same skill as in *Urban Heat Islands* and *Air Quality*.
- **Environmental education** — climate change, with an honest distinction between what a jar

shows and what the atmosphere does.

The core skill is **attributing a cause**. Anyone can see that the jar warmed up. The classroom question — and the upper-secondary skill — is *how do we know what warmed it.*

3. Time plan

Route	Time	What happens
Demonstration	**20 min**	One closed container, the lamp on, the three cards watched together. **It ends with the question, not with the conclusion**: what warmed it? The list of possibilities goes on the board.
Full, lesson 1	**50 min**	The written hypothesis about the cause (5), setting up and checking the lamps with the light sensor (5), the scatter of the two empty jars: ten minutes of settling and five of readings (15), the jar with a lid against the one without, minute by minute (20), the difference and the verdict (5).
Full, lesson 2	**50 min**	Raising the CO_2 in one jar and the same gesture in the control (5), ten minutes of settling (10), twenty-five minutes of minute-by-minute measurement (25), the difference, the comparison with the threshold from lesson 1 and the verdict (10).
Extension	in lesson 2	A third container, **without a lid**, alongside the other two. It separates the lid's effect from everything else, in the same run.

4. Materials

Item	Per class	Notes
MagicBox units with SCD41 and the light sensor	2	**Two, in practice compulsory.** With one there is no control, and without a control the activity's question has no answer.
Laptop with the app	1	**Two windows** are opened, one per box.
Identical transparent containers with lids	2–3	**Identical**, compulsory: the same material, the same volume, the same wall thickness. The third, without a lid, is for the extension.
Two identical lamps, or one lamp midway	1 set	If you use a single lamp it stands exactly midway, at measured equal distances — and check the equality *with the light sensor*, not by eye.
Bicarbonate of soda and vinegar	a little	For raising the CO_2 chemically, **in a separate bottle**, from which the gas is poured into the jar — as the app proposes. The alternative: breathing gently into the container (see below what each spoils).
Two identical empty bottles, 0.5–1 l	2	The reaction happens in one; from the other, empty, air is "poured" into the control, so the gesture is the same.
Ruler	1	For the lamp-to-jar distances, which must be measured, not judged.
Stopwatch	1	Readings are taken at fixed intervals.

Preparation. Choose containers that really are identical — two jars of the same make, not "two jars". Set up the lamps and **check with the light sensor that the illuminance is the same in both**; if they differ by more than a few per cent, move the lamp until they match, because otherwise you have changed two factors. **Do the trial run with both jars empty**, ten minutes under the lamps: the temperature difference that appears there, with nothing changed, is the scatter of the rig and the threshold below which nothing can be claimed.

5. Scientific background for the teacher

What actually warms a jar under a lamp

Several mechanisms can warm the air in the container, and they are not equal.

Convection being stopped — by far the dominant effect. Air warmed by the lamp and the walls would rise and leave. The lid stops it. This is the mechanism of a garden greenhouse and of a car left in the sun, and it produces warmings of several degrees within minutes.

Absorption in the walls and contents. Glass and plastic absorb part of the light and warm up; they then warm the air by conduction. A darker or thicker container warms differently from a thin, clear one — *which is why the containers must be identical.*

The box's own heat. The circuit board in the box draws current and warms a little. In a closed jar it is a small heat source — which is why the electronics stay outside as far as possible, and each jar holds a box of the same kind.

Infrared absorption by CO_2 — the effect being sought, and the smallest of all. In a volume of a few litres the optical path is of the order of tens of centimetres, and the amount of CO_2 along it is small. The radiative effect exists, but it is far below the other two and **can be completely hidden by them**.

The consequence, which is the lesson: **a single jar warming up proves nothing about CO_2.** Only the difference between two identical jars, treated identically, differing *only* in their CO_2 content, can be attributed to the gas — and only if it exceeds the measured scatter of the rig.

5.1 Why a garden greenhouse is not a good model for the atmosphere

A garden greenhouse works by **stopping convection**: the glass keeps the warm air inside. The atmosphere has no lid and does not stop convection — on the contrary, convection is the engine of the weather.

The atmospheric greenhouse effect is **radiative**: the Earth's surface emits infrared, and greenhouse gases **absorb part of it and re-emit it** in all directions, including back towards the surface. They do not "hold the heat" or "block" the radiation: they absorb it and send it on, partly back down. The result is that *the altitude from which radiation effectively escapes to space* rises, and the surface settles at a higher temperature.

The name is a historical accident, and the analogy misleads at exactly the point that matters. Say so explicitly to the class; upper-secondary pupils retain it, and lower-secondary ones retain at least that "it is not like a greenhouse".

5.2 What this activity really does prove — and it is better

The activity cannot convincingly demonstrate radiative warming in a jar. It can, however, demonstrate something better, with no extra apparatus: **that carbon dioxide absorbs infrared radiation**.

The proof is in the pupils' own hands. **The SCD41 is an NDIR instrument**: it measures CO_2 by sending infrared at 4.26 micrometres through a small chamber and seeing how much reaches

the detector. The more CO_2, the less arrives.

Therefore: *the fact that the sensor can measure CO_2 is itself the proof that CO_2 absorbs infrared* — which is exactly the property that makes it a greenhouse gas. **The number on the screen is not a hint of the greenhouse effect; it is a direct consequence of its mechanism.**

This is a link pupils can verify by reasoning rather than merely accept, and it deserves more class time than trying to extract a statistically doubtful difference between two jars.

5.3 Raising the CO_2: the two methods and what each spoils

Breathing into the container is the simplest and raises the CO_2 to a few thousand ppm. **But exhaled air is both warm and humid**: it raises the starting temperature and the humidity, so it changes three things at once — and water vapour is itself a greenhouse gas that absorbs infrared, so you no longer even know *which* gas would have warmed the jar. If you breathe, **wait ten minutes** for the temperature to come back before starting, and breathe the *same amount of air* into the control — otherwise the control is no longer a control.

Bicarbonate and vinegar produce CO_2 without heating and easily raise it above 10 000 ppm. **But the reaction absorbs heat** — it is mildly endothermic, so a cup placed *inside* the jar *cools* it at first — and produces water, while the vinegar also gives off acetic-acid vapour, so it changes both the humidity and the air's make-up. That is why the app proposes the clean version: **the reaction happens in a separate bottle**, and the gas — heavier than air — **is poured** slowly into the jar, like water. Into the control, with the same gesture, air is "poured" from an identical empty bottle. Then **ten minutes of settling with the lamp off**, before switching it on.

Neither method is clean, and saying so to the class is worth more than picking one and staying quiet. **The cleanest method is in fact a third**: do not raise the CO_2 at all, but compare a jar with a lid against one without — and so measure the convection effect, which is the dominant and real mechanism.

6. Ready-made investigations

6.1 Investigation 1: the scatter of the rig — first, always

Research question. How much do our two jars differ when we have changed nothing?

Method. Both containers empty — each holding only its box —, closed, under identical lamps, for fifteen minutes: ten of settling and five of readings, every minute.

What to measure. The largest temperature difference between the two jars. **That is the scatter of the rig.**

Expected finding. Typically 0.2–1.5 °C, depending on how identical the containers and lamps are. Pupils then write the sentence that governs the rest: *we will treat a difference between the jars as real only if it exceeds _____ °C.*

6.2 Investigation 2: the lid — the dominant mechanism

Research question. How much of the warming is simply convection being stopped?

Method. Two identical containers, both with ordinary air, under identical lamps: one **with a lid**, the other **without**. Twenty minutes, with readings every minute.

What to measure. The temperature in each, and the difference between them.

Expected finding. **A large and unambiguous difference**, typically 2–6 °C after twenty minutes — far above the scatter of the rig. This is *the garden greenhouse*, measured. And it is the most certain piece of evidence the activity can produce.

6.3 Investigation 3: CO_2 against a control — the hard experiment

Research question. Does the jar with raised CO_2 warm more than the control?

Method. Both containers **with lids**, identical, under identical lamps. In one the CO_2 is raised; in the other the *same gesture* is made without the gas — air poured from an empty bottle instead of CO_2 poured from the reaction bottle, or the same amount of breath —, then ten minutes of settling with the lamp off, in both. The lamps are switched on and twenty-five minutes of measurement follow.

What to measure. The temperature difference between the jars, compared with the scatter from investigation 1.

Expected finding. Most likely, a difference comparable with the scatter — that is, *we cannot detect it.* Sometimes, with large containers and very high CO_2, a small difference of the order of 0.5–1 °C appears.

A negative result is the correct result here, and must be treated as such, not as a failure. The conclusion to write: *at the scale of our jar, the radiative effect of CO_2 is too small for us to distinguish from the scatter of the rig — which does not mean it does not exist, but that our rig cannot measure it.* **That is a researcher's sentence.**

The investigation, in one table

The question. What warmed the jar?
The hypothesis. Written **beforehand**: the lid, the colour of the walls, the CO_2, or something else — and what temperature difference each would be expected to give. Not changed afterwards.
What I change. One thing at a time: first the lid (investigation 2), only then the CO_2 (investigation 3). Anyone who changes both at once can attribute nothing.
What I measure. The temperature in each container, every minute, for the whole series — **and the illuminance in each**, as evidence that the lamps are equal.
What I keep the same. Identical containers, lamps at measured equal distances, the same starting temperature, the same starting humidity, the same position of the box inside, and **the same gesture made in both jars** — if I pour gas from a bottle into one, I pour air from an identical bottle into the other.
How many repeats. The series is repeated with **the jars swapped**. If the difference follows the *treatment* and not the *jar*, it is real; if it stays with the same jar, you have a container difference, not a gas one.
What makes a reading valid — and what makes it an artefact. Valid is a reading taken after ten minutes of settling, with the illuminance verified equal and the difference compared against the rig's scatter. An artefact is anything else: **a jar two centimetres closer to the lamp warms measurably more**; exhaled air brings heat and moisture along with the gas; a hand resting on a lid warms it; and a difference smaller than the scatter does not mean "CO_2 does not warm" but **"we cannot detect it"**.

7. Analysis after the activity

7.1 What the data should look like

Situation	Typical value	What it means
Scatter of the rig, empty jars	0.2–1.5 °C	**The threshold below which nothing can be claimed.**
With lid against without, 20 min	+2 to +6 °C	**Convection stopped.** The dominant mechanism, unambiguously.
Raised CO_2 against control, 25 min	0 to +1 °C	Usually *below or at* the scatter. A negative result, and a correct one.
CO_2 after breathing in	2000–6000 ppm	But temperature and humidity jump with it.
CO_2 poured from the bicarbonate-and-vinegar bottle	5000 to above 10 000 ppm	No temperature jump. With the cup placed *inside* the jar, a slight cooling and extra water and vinegar vapour.
Illuminance in the two jars	must differ by under 5%	If it differs more, you have changed two factors.
Starting temperature, both	must differ by under 0.5 °C	Otherwise the series starts crooked.

7.2 Questions to ask afterwards

1. The jar with a lid warmed far more than the one without. Which of the three modes of heat transfer explains that, and why not the other two?
2. The difference between the CO_2 jar and the control was smaller than the scatter of the rig. What can and cannot you conclude from that?
3. A garden greenhouse and the atmosphere warm by different mechanisms. Describe each, and say why the name "greenhouse effect" is a poor one.
4. Your sensor measures CO_2 by sending infrared through a chamber. What does the very fact that it works prove about the properties of CO_2?
5. How would you design an experiment that *could* detect the radiative effect? What would have to be different — larger, longer, more precise?

7.3 Assessment ideas

- **Low stakes:** the pupil records the two temperature series and says which jar warmed more, and by how much.
- **Standard:** the pupil compares the difference with the rig's scatter and writes the correct verdict of three — detected, borderline, not detected — for each pair.
- **Extension:** a one-page piece distinguishing the garden greenhouse mechanism from the atmospheric one, explaining why the rig cannot detect the second, and proposing what would have to change so that it could. The strongest pupils use **the NDIR sensor's own operation** as independent evidence that CO_2 absorbs infrared.

Common pitfalls and misconceptions

"The jar warmed up, so we have demonstrated the greenhouse effect." The central error of this topic. It warmed up because it has a lid. Without a control, nothing can be attributed to the gas.

A single jar. Without a control the activity can produce a demonstration but not a

conclusion. If you have one box, run the demonstration and **stop at the question**, not at the answer.
Changing two things at once. The lid *and* the CO_2 in the same step. Do the lid first, on its own.
"Nearly identical" containers. Different wall thicknesses and colours produce differences of their own larger than the effect being sought.
Unverified lamps. Check the equality with the light sensor, not by eye: two "identical" lamps can give illuminances differing by tens of per cent.
Breathing only into the test jar. It brings heat and moisture along with the CO_2, and into only one of them — and water vapour absorbs infrared too. If you breathe, breathe into both, and wait.
Electronics in the jar. A power bank or a second device inside the jar warms it more than any gas. Only the box goes in, and a box of the same kind in the control.
"CO_2 holds the heat." The correct phrasing: greenhouse gases *absorb* infrared and *re-emit* it, partly back towards the surface.
Reading after two minutes. The differences pursued are small and slow. Ten minutes of settling, then at least twenty minutes of measurement.
Treating a negative result as a failure. *We cannot detect it* is a result, and here it is the most likely correct one.

Safety and classroom management

Never a flame in a closed container — the warning is on the app's screen too, and it is the most important one in this activity. No candle, no match, nothing burning. A closed container with a flame inside can fail, and the consumption of oxygen makes the experiment both dangerous and meaningless.
Lamps get hot, particularly filament ones, and here they stand beside glass and plastic lids. Do not cover them, keep them at a measured distance, and let them cool before packing away. A glass jar heated unevenly can crack: prefer plastic, or keep the lamp no closer than 20 cm.
Bicarbonate and vinegar are safe, but the reaction foams and can overflow. Do it in the separate bottle, with only a little vinegar, on a tray, and pour only the gas into the jar, never the liquid. **The liquid never touches the MagicBox**: the box is not waterproof.
Do not stopper the bottle in which gas is being produced, and do not seal the jars airtight: the pressure rises, and a stopper or a lid screwed down hard can fly off. A lid resting in place, not screwed tight, is enough for the experiment.
Hygiene when breathing in — if you choose the exhaled-air method, each pupil breathes into their own container, or the teacher does it.

Adaptations for pupils with special educational needs

Two series of numbers, one graph. The main adaptation: the pupil is given a sheet with the axes already drawn and *one* curve already plotted — the control's — and draws only the second over it. **The gap between the curves is the result**, and it is visible with no arithmetic at all.
The threshold, written once and large. The rig's scatter is written once, large, at the head of the sheet, and every comparison refers to it. The question becomes binary.
Three real roles for pupils with motor difficulties. The rig needs handling, but the roles that matter are done from a desk:

- **the symmetry keeper** — checks before the start that the illuminance and the temperature are equal in both jars, and has the right to stop the series if they are not; this is the role with the greatest scientific value, because they defend the very idea of a control;

- **the timekeeper** — announces the minute-by-minute readings;
- **the scribe** — writes the two series side by side and computes the difference as it arrives.

Sensitivity to light and heat. Bright lamps and warm jars can be unpleasant; all three roles are done from a distance.

If things do not go to plan

You have only one box. Run **investigation 2** — lid against no lid — measuring in turn, under the same conditions, ten minutes each. You lose simultaneity, but the lid's effect is so large that it shows anyway. **Do not attempt investigation 3 with one box**: the effect is far too small for a sequential measurement.
You have no identical containers. Use *the same* container for both conditions, in turn, with the same lamp and the same duration. It is slower, but it removes the container difference — the largest source of error.
You have no second lamp. One lamp, exactly midway, with measured equal distances **verified with the light sensor**.
The difference comes out smaller than the scatter. That is the expected result. Write it as such and move on to the best part of the lesson: why the sensor itself proves that CO_2 absorbs infrared.
The temperature jumps when you breathe in. Naturally — exhaled air is at 35 °C. Wait ten minutes for it to come back, and breathe into the control too.
The box will not connect. Do not stop the lesson. Use the "What the data should look like" table as a printed dataset: pupils are given the scatter and the two series, compute the difference, compare it with the threshold and write the verdict. **The whole business of attributing a cause — which is the lesson — is done without the equipment.** Measure again another day, after checking the cable, the port selected in the app and, if you work over WiFi, the network.

8. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are base addresses, not deep links, so they do not rot.

- **NASA Climate** — `https://climate.nasa.gov` — search for the greenhouse effect and the Earth's energy budget; the radiative explanation, correctly framed.
- **PhET Interactive Simulations** — `https://phet.colorado.edu` — search for the greenhouse effect simulation; **it shows the radiative mechanism the jar cannot**, and it is the natural complement to this activity.
- **Institute of Physics** — `https://www.iop.org` — search for material on heat transfer and radiation; useful for investigation 2 and for the discussion of convection.
- **Sensirion** — `https://sensirion.com` — the SCD4x datasheet, if you want to show the class that the NDIR principle is precisely infrared absorption by CO_2.
- **Romanian National Meteorological Administration** — `https://www.meteoromania.ro` — for local climate data, if the discussion turns to observed changes in Romania.
- **IPCC** — `https://www.ipcc.ch` — the synthesis reports, for upper-secondary pupils who want the primary source for claims about climate.

9. Assessment

The exit ticket, identical to the one on the pupil sheet: *what warmed the jar, and how do you know?* Collected in the last two minutes, it says more about the class's understanding than any mark.

For a more formal judgement, use the **common three-axis rubric** — *measurement practice*, *data interpretation* and *scientific reasoning*. At Explorer and Scientist level the three axes look like this:

- **Measurement practice** — the pupil measures the rig's scatter before anything else, verifies the lamps' equality with the sensor, changes one thing at a time, and makes the same gesture in both jars.
- **Data interpretation** — the pupil compares the difference with the scatter and uses three verdicts, not two; **they treat a negative result as a result**.
- **Scientific reasoning** — the pupil distinguishes the garden greenhouse mechanism from the atmospheric one, explains why the rig cannot detect the second, and uses the NDIR sensor's operation as evidence that CO_2 absorbs infrared.

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

10. Pupil sheet

The sheet the pupils fill in is a separate document: `docs/en/student_sheets/greenhouse_effect.pdf`. It has two pages — the scatter, the hypotheses about cause and the lid experiment on the first; the CO_2 experiment, the comparison with the threshold and the conclusion on the second — so it is printed double-sided, one sheet per pupil. It is built on the three layers — what I see on the screen, what I do on my own, the conclusion I write.

Three things to know before printing it. **The sheet asks for a hypothesis about the *cause*, not about the result**: at the start, the pupil ticks which mechanism they think will warm the jar. **The rig's scatter has a large box of its own at the head of the page**, and every comparison refers to it. **The verdict "not detected" is printed as an option on equal footing with the other two**, precisely because in the CO_2 investigation it is the most likely result — and a sheet that left no room for it would push pupils into reporting an effect they did not measure. The tables are not repeated here precisely so that the two documents cannot drift apart: there is one printable version, and it is the sheet.